



Highly selective Protein A resin allows for mild sodium chloride-mediated elution of antibodies

Julia Scheffel, Sophia Hober*

Department of Protein Science, KTH-Royal Institute of Technology, SE-10691, Stockholm, Sweden

ARTICLE INFO

Article history:

Received 2 November 2020

Revised 11 December 2020

Accepted 22 December 2020

Available online 24 December 2020

Keywords:

Affinity chromatography

Protein A

Antibody purification

Calcium-dependent

Z_{Ca}

Elution

ABSTRACT

The manufacturability of therapeutic monoclonal antibodies is limited by the harsh conditions that antibodies are subjected to during the purification procedure, which in turn restricts the development of novel acid-sensitive antibodies. The gold standard for antibody purification, Protein A affinity chromatography, offers the selective capture of antibodies with great yields, but also poses a threat to the quality of the antibodies. Antibodies and Fc-fusion proteins risk forming aggregates as a consequence of the acidic elution from the Protein A ligands, compromising the potency and safety of the drug. Here, we present a novel, mild purification strategy based on a calcium-dependent ligand derived from Protein A, called Z_{Ca}. Antibodies captured on a high-capacity tetrameric Z_{Ca} resin in the presence of calcium can be eluted by removing the calcium ions through the addition of a chelator, and we describe the strive to find a sustainable alternative to the previously applied chelator EDTA. The naturally occurring chelator citrate is shown to seamlessly replace EDTA. Further buffer optimization reveals that the elution can be considerably improved by increasing the conductivity through the addition of 300 mM sodium chloride, leading to a very concentrated eluate. Remarkably, merely sodium chloride at a concentration of 50 mM is proven to be sufficient for calcium-dependent antibody release in a cost-efficient manner. Antibodies of subclasses IgG2 and IgG4 are eluted with sodium chloride at neutral pH and IgG1 at pH 6, due to varying affinities for the tetrameric Z_{Ca}, ranging between 90–780 nM. The mild elution of an IgG4 antibody eliminated the formation of aggregates, which constituted as much as 34% of all eluted antibody from MabSelect SuRe at pH 3. This novel purification strategy thus combines the valuable qualities of a Protein A resin, by providing high selectivity and a recovery of 88–99%, with an exceptionally mild elution step similar to ion-exchange chromatography, rendering considerably more functional antibody.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>)

1. Introduction

The therapeutic monoclonal antibody (mAb) field is ever-expanding [1], and efficient methods able to selectively purify a wide variety of antibodies are crucial for antibody manufacturability and continued innovation. Today, there are numerous separation techniques available for the isolation of antibodies as well as Fc-fusion proteins from impurities produced upstream [2]. Among the less prominent approaches are non-chromatographic separation methods, such as precipitation [3] and aqueous two-phase extraction [4], which have not gained widespread use, unlike chromatographic techniques. The high selectivity and recovery of Protein A affinity chromatography and the high capacity of ion-exchange chromatography (IEX) are some of the aspects which

have rendered these two of the most widely used chromatography techniques. IEX also provides mild separation conditions by increasing the ionic strength to release the antibodies from the column, as opposed to Protein A affinity chromatography: the most well-established method for antibody purification [5]. For elution of antibodies captured on a Protein A resin, the pH is shifted from neutral to low pH of 3–3.5, contributing to antibody aggregation to different extents depending on the antibody. To remove these aggregated species, which are classified as impurities that risk eliciting an immunogenic response, IEX is often included further downstream in a multistep purification procedure [6–10].

However, the formation and removal of antibody aggregates compromise the yield of the purification process, and some antibodies completely degrade under these acidic conditions [10]. In order not to impede the development of such antibodies or Fc-fusion proteins that show promising therapeutic potential, there have been repeated attempts to milder the elution of the oth-

* Corresponding author.

E-mail address: sophia@kth.se (S. Hober).

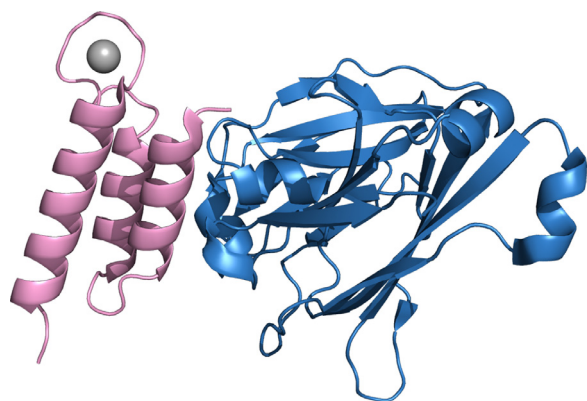


Fig. 1. Crystal structure of Z_{Ca} in pink in complex with Fc of human IgG1 in blue (PDB ID 6FGO [16]). The introduced loop binds to calcium (gray) and the domain interacts with Fc by helix one and two.

erwise successful Protein A affinity chromatography. Studies have been published regarding the role of buffering agents and additives, some which allowed for a slight increase in the elution pH and some which had varying effects on the aggregation propensity depending on the antibody to be purified [11]. Furthermore, engineering of Protein A ligands, by incorporating binding site mutations [12] and loop extension [13], has also enabled the elution of antibodies at slightly higher pH, but still far from neutral conditions. Engineering by introducing mutations aimed to decrease the hydrophobicity and the ligand stability has rendered a thermosensitive Protein A domain, from which antibodies can be eluted by raising the temperature from 5°C to 40°C [14]. The antibody binding of this domain, also known as Byzen Pro, was later proven to be salt-dependent [15]. A very high concentration of sodium chloride, 1.5 M, can be used for elution at neutral pH, but with suboptimal binding capacities at room temperature.

We have recently reported a novel engineering approach to overcome the acidic elution, namely the design of a Protein A-based ligand with a calcium-dependent binding to antibodies [16]. In the Z domain, derived from the B domain of Protein A, the original loop between the second and third alpha helix was exchanged to a calcium-binding loop, based on one of the loops of calmodulin. Within the loop and in close vicinity of it, a number of randomizations were introduced to create a phage library. Through phage display selections from this library and a subsequent library based on one of the selected variants, a purification ligand comprising a calcium-dependent affinity to antibodies could be isolated. The structure of the ligand, Z_{Ca}, in complex with Fc of IgG1 is displayed in Fig. 1. From a matrix based on this ligand, antibodies can be eluted with the calcium chelator ethylenediaminetetraacetic acid (EDTA) at pH 5.5–7, depending on the subclass, leading to an equally concentrated elution pool as for acidic elution from commercial Protein A resins [17]. Moreover, a chromatography resin coupled with a tetrameric version of Z_{Ca}, denoted Z_{Ca}TetraCys, provides a dynamic binding capacity at the same level as the commonly used MabSelect SuRe. Z_{Ca}TetraCys also maintains the high selectivity of Protein A, providing a reduction of host cell proteins in the same range as the newly launched MabSelect PrismA.

Nevertheless, the use of EDTA in a large-scale purification process is problematic since EDTA has been prohibited for industrial use in several countries [18]. The chelator shows poor biodegradability, reduces the clearance of heavy metals from wastewater, and remains in high concentrations after wastewater treatment, which combined with its ability to mobilize heavy metals in soil could lead to potentially harmful effects on the environment [19]. In this study, we evaluated if EDTA can be replaced with other agents while maintaining the mild elution from the Z_{Ca}TetraCys column.

A more readily degradable alternative is the chelating agent citrate [19], which is also frequently used as an elution agent at low pH in conventional Protein A-based purification. It was investigated if the chelating properties of citrate were strong enough to fully elute all antibody that was captured on Z_{Ca}TetraCys in the presence of calcium. The effect of adding sodium chloride to the elution buffer was studied, including the calcium-dependency of the elution. Furthermore, the optimal elution properties to achieve a concentrated product at mild conditions were determined for each IgG subclass, and related to their respective affinities for Z_{Ca}TetraCys. The selectivity and recovery of the resin was evaluated, as well as the ability to prevent the formation of antibody aggregates. Here, we present a high-capacity purification strategy that combines the antibody specificity of Protein A affinity ligands with mild elution comparable to ion-exchange chromatography. The proposed method allows for the selective and efficient purification of sensitive or aggregation-prone antibodies or Fc-fusion proteins.

2. Materials and methods

2.1. Monoclonal antibodies and sample preparation

IgG1 (trastuzumab) expressed in a Chinese Hamster Ovary cell line (CHO-M D1E7) at a concentration of 0.8 g/L was provided by Cytiva. CHO cell supernatant (produced in-house) was spiked with IgG4 (BioInvent) to a concentration of 0.1 g/L. The CHO cell supernatants were stored at +4°C. Prior to preparative affinity chromatography, the supernatants were filtered using 0.2 µm filters (Sarstedt, 83.1826.001) and 1 mM CaCl₂, or 10 mM CaCl₂ for the selectivity purifications, was added to promote binding to Z_{Ca}TetraCys. Pure IgG2 (Department of Biotechnology, University of Natural Resources and Life Sciences, Vienna) and IgG4 (BioInvent) was diluted in 1xTBS (50 mM Tris, 150 mM NaCl), 1 mM CaCl₂, pH 7.5 to reach a final concentration of 1.2 mg/ml. For recovery analysis, pure IgG1 (rituximab, Roche, 494237) and IgG4 (BioInvent) was diluted to 2 mg/ml in 1xTBS, 10 mM CaCl₂. For capture and elution on MabSelect SuRe as a control, pure IgG4 was diluted in 1xPBS (20 mM sodium phosphate, 150 mM NaCl), pH 7.2.

2.2. Chromatographic equipment

All affinity chromatography was conducted on an ÄKTA Pure chromatography system (Cytiva) at 25°C. UV was measured at 280 nm. The UNICORN 6.4 software (Cytiva) was used for evaluation of the resulting chromatograms. CIP was regularly performed on the system, while the columns were detached.

Size-exclusion chromatography (SEC) was performed on an NGC Chromatography System (BioRad) at 25°C. UV was measured at 280 nm. For evaluation of the runs, the ChromLab software (BioRad) was used.

2.3. Stationary phase and buffers

In this study, two 1 ml HiTrap columns containing an agarose-based resin with immobilized Z_{Ca}TetraCys ligands were used; one column for all preparative purifications and one column for all eluent evaluation studies. Z_{Ca}TetraCys was previously cloned, produced, purified and coupled to the chromatography resins as published in Scheffel et al [17]. Site-specific immobilization was achieved by coupling of the unique C-terminal cysteine of Z_{Ca}TetraCys to the resin through thiol-based chemistry. For comparison to a commercial Protein A resin, a 1 ml HiTrap MabSelect SuRe column was used. A Superdex 200 Increase 5/150 GL column (Cytiva) was used for SEC analysis.

The buffers used in affinity chromatography for the eluent evaluation on Z_{Ca}TetraCys include two wash buffers: 1xTBS, 0.05% (v/v)

Tween20, 1 mM CaCl_2 , pH 7.5 (wash buffer 1) and 5 mM NH_4Ac , 1 mM CaCl_2 , pH 5–6 (wash buffer 2). The elution buffer used in the preparative purifications consisted of 100 mM NH_4Ac , 100 mM EDTA, pH 5.5 (elution buffer) while the buffers used for eluent evaluation are listed in Table S1. As a control eluent after each evaluative purification, 0.3 M HAc, pH 3.3 (control elution buffer) was applied. The columns were stored in 1xTBS, 0.05% (v/v) Tween20, 1 mM CaCl_2 , and 20% Ethanol (storage buffer). The buffers used for capture and elution on MabSelect SuRe included 1xPBS, pH 7.2 as binding buffer and 0.1 M sodium citrate, pH 3.0 as elution buffer. For SEC experiments, 1xPBS was used as running buffer. For cleaning-in-place (CIP) of the systems, 0.5 M NaOH was used. All buffers were filtered using 0.45 μm filters (Sartorius, 11105–47—N) and degassed before use.

2.4. Antibody purification

A flow rate of 1 ml/min was consistently used, unless otherwise stated. IgG1 was purified from the CHO cell supernatant for use in the eluent evaluation. The Z_{Ca} TetraCys column was first pulsed with 6 column volumes (c.v.) of wash buffer 1 and 6 c.v. of elution buffer, followed by equilibration with 13 c.v. wash buffer 1. Next, ca 40 ml of the IgG1-containing supernatant was applied onto Z_{Ca} TetraCys at a flow rate of 0.5 ml/min. The column was washed with 13 c.v. wash buffer 1 and 5 c.v. wash buffer 2 before elution with 6 c.v. elution buffer. The eluate was fractionated and the 5 fractions ($5 \times 500 \mu\text{l}$) with the highest absorbance, according to the chromatograms, were collected and buffer exchanged to 1xTBS, 1 mM CaCl_2 for subsequent use. Regeneration of Z_{Ca} TetraCys was conducted using 3 c.v. elution buffer, 10 c.v. control elution buffer, 6 c.v. wash buffer 1, 5 c.v. deionized water and 2 c.v. storage buffer.

Different eluents (Table S1) were evaluated for elution of captured IgG1, IgG2 or IgG4 from Z_{Ca} TetraCys. Approximately 2.4 mg of pure antibody, insufficient to saturate the column, was injected at 0.5 ml/min through a 2 ml-sample loop by 4 c.v. of wash buffer 1, to achieve comparable peak areas under the elution curves. The same buffers and c.v. as previously described were used here, with the exception of the eluents, which were applied in a gradient or isocratic elution for 20 c.v. (Table S1). In addition, a second elution using 10 c.v. control elution buffer was conducted after each run to check for residual antibody. Capture and elution of IgG4 (BioInvent) on MabSelect SuRe was conducted according to the application note from the manufacturer.

Purifications to determine the selectivity and recovery of Z_{Ca} TetraCys were conducted as described, but with 10 mM CaCl_2 in wash buffer 1, excluding washing with wash buffer 2. CHO cell supernatant containing trastuzumab (Cytiva), CHO cell supernatant (produced in-house) spiked with IgG4 (BioInvent), pure IgG1 (rituximab, Roche, 494237) or pure IgG4 (BioInvent) were applied to Z_{Ca} TetraCys at a volume of 2 ml by 6 c.v. of wash buffer 1. IgG1 and IgG4 were eluted with 20 c.v. of 100 mM NH_4Ac , 300 mM NaCl, pH 5.5 and 20 mM HEPES, 300 mM NaCl, pH 7, respectively. All steps of the purification procedure were fractionated and pooled individually. The absorbance at 280 nm was measured to determine the protein content.

2.5. Surface plasmon resonance

IgG1, IgG2 and IgG4 were immobilized on a CM5 chip to approximately 300 RU via amine coupling. A Biacore 8K instrument (Cytiva) was used for analysis with 1xHBS (20 mM HEPES, 150 mM NaCl), 0.05% (v/v) Tween20, 1 mM CaCl_2 as running buffer. The protein Z_{Ca} TetraCys was diluted to 2, 8, 32, 128 and 512 nM in running buffer and single-cycle kinetics were run at 30 $\mu\text{l}/\text{min}$. The association time was 240 s with a final dissociation of 600 s with running buffer, followed by 60 s of surface regeneration using 10

mM HCl. Data were fitted with a Bivalent analyte model to determine the kinetic parameters, using the Biacore Insight Evaluation Software (Cytiva).

2.6. SDS-PAGE

The purity of the fractions throughout the antibody purification was determined by SDS-PAGE. Equal fractions in relation to the full volume of each sample from each purification were mixed with a reducing buffer to a concentration of 30 mM Tris-HCl, 1% SDS, 0.01% bromophenol blue, 5 mM TCEP and 13% glycerol. The samples were heated to 95°C for 5 min, loaded on a NuPAGE Novex 4–12% Bis-Tris gel (Invitrogen, NP0323BOX), including a PageRuler™ Plus Prestained Protein Ladder (Thermo Scientific, 26620), and run in 1xMES (50 mM MES, 50 mM TRIS, 1 mM EDTA, 0.1% SDS, pH 7.3) running buffer at 180 V, 4°C for 50 min. The gel was washed in deionized water for 3×5 min, after which GelCode® Blue Safe Protein Stain (Thermo Scientific, 24594) was added for staining during 1 hour, and then destained in deionized water ON.

2.7. Size-exclusion chromatography

The proportions of high-molecular weight species formed by elution from Z_{Ca} TetraCys and MabSelect SuRe were investigated by SEC. The eluates obtained from capture and elution of pure IgG4 antibody (BioInvent) on the two columns were immediately buffer exchanged to 1xPBS. 25 μl of the buffer exchanged eluates were injected onto the Superdex 200 Increase 5/150 GL column (Cytiva) at a concentration of 0.8 mg/ml. A flow rate of 0.45 ml/min was used, and the absorbance was measured at 280 nm. A calibrant was also run which included aldolase, which has a protein molecular weight of 158 kDa.

3. Results and Discussion

3.1. EDTA can be replaced by citrate for mild antibody elution

In previous studies, we have demonstrated efficient elution of IgG captured on a Z_{Ca} TetraCys column by the chelating agent EDTA at pH 5.5 [16,17]. Due to the strive to phase out EDTA from the industry, more environmentally friendly chelators were considered for the removal of calcium and subsequent release of the bound antibodies. The chelator ethylene glycol-bis(β -aminoethyl ether)-N,N,N',N'-tetraacetic acid (EGTA) has previously been shown to elute antibodies from a Z_{Ca} monomer column, but with a suboptimal elution profile compared to EDTA. As it is also significantly more expensive than EDTA, it was concluded to be disadvantageous, but still supported the notion that EDTA can be substituted [17]. The present study investigated whether the naturally occurring chelator citrate could be a viable alternative to the synthesized EDTA for mild elution of antibodies. Citrate, which is fully biodegradable, is also a commonly used eluent in conventional acidic elution from Protein A resins. The chelator was evaluated for elution of human IgG1 (trastuzumab), which had been captured on Z_{Ca} TetraCys. The elution was conducted at pH 5.5, based on previous studies with EDTA, in the absence of sodium chloride. This resulted in a strikingly similar elution curve to EDTA at the same eluent concentration (Fig. 2). The elution of the antibody started at the same concentration of each chelator, and citrate-mediated elution generated an equally concentrated antibody product as for EDTA-mediated elution. Thus, citrate proved equal to EDTA in its ability to chelate calcium and elute captured trastuzumab from Z_{Ca} TetraCys at pH 5.5. This despite the fact that citrate has a lower binding affinity towards calcium, and could consequently require the use of higher concentrations to chelate the same number of calcium ions [20]. Citrate even showed superior elution properties

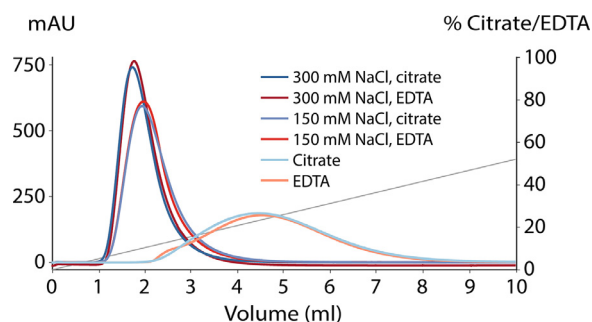


Fig. 2. Comparison of EDTA and citrate as elution agents. Trastuzumab, loaded onto Z_{Ca} TetraCys in similar amounts for each chromatographic run, eluted at pH 5.5 with a gradient (gray) of 0–100 mM EDTA (red shades) or citrate (blue shades), including the addition of 150 mM or 300 mM sodium chloride. Citrate-mediated elution was shown to be equivalent to that of EDTA, and the elution was enhanced by the use of sodium chloride in the eluent.

over EDTA as it allowed for the complete elution of trastuzumab up to pH 6 (Fig. S1), an increase in pH by a factor of 3.2 compared to EDTA-mediated elution requiring pH 5.5. However, the elution pool at pH 6 is twice as diluted as the elution pool at pH 5.5 (Table S2), and a further increase in pH led to small amounts of antibody retained on the column after elution (Fig. S1).

Increasing the ionic strength of the elution buffer, by the addition of sodium chloride, has previously been shown to improve the elution for a polyclonal antibody sample captured on Z_{Ca} TetraCys by resulting in a sharper peak [17]. Here, we studied the effect of sodium chloride on both EDTA- and citrate-mediated elution for a monoclonal antibody. Capture of equal amounts of trastuzumab on Z_{Ca} TetraCys was followed by elution with a gradient of citrate or EDTA in combination with different concentrations of sodium chloride. Elution with 150 mM sodium chloride differed greatly from elution with the chelators alone, while 300 mM sodium chloride further sharpened the elution peak, leading to a highly concentrated eluate (Fig. 2). Thus, sodium chloride appears to play an important role in the elution of antibodies from Z_{Ca} TetraCys. Moreover, EDTA- and citrate-mediated elution responded similarly to the addition of sodium chloride in the case of trastuzumab. Altogether, the findings presented suggest that EDTA is replaceable by means of citrate-mediated elution. In addition to the environmental benefits associated with the substitution of EDTA with citrate, the chelator is also favorable from an economic perspective since it is considerably less expensive than EDTA.

3.2. Sodium chloride alone provides efficient elution of all captured antibody

Since the combination of sodium chloride with citrate generated comparable elution profiles to sodium chloride and EDTA, the

former combination was further evaluated. For the purpose of establishing the optimal concentration of citrate and sodium chloride in the eluent, lower concentrations of citrate were first evaluated. Accordingly, citrate gradients with decreasing final concentrations were examined for elution of trastuzumab with a constant sodium chloride concentration of 300 mM at pH 5.5. Despite the decrease in citrate concentrations, the chromatograms demonstrate uniform elution profiles (Fig. 3A). In fact, the elution progressed in the same manner in the complete absence of citrate in the eluent, indicating that citrate is not responsible for elution at high sodium chloride concentrations at pH 5.5. Instead, 300 mM sodium chloride without citrate led to the complete elution of all antibody captured on Z_{Ca} TetraCys, into a very concentrated product.

With a new alternative eluent, elution closer to neutral pH is of interest for minimal impact on the captured mAbs or Fc-fusion proteins. Thus, sodium chloride was studied at higher pH than 5.5 for elution of trastuzumab from Z_{Ca} TetraCys. Remarkably, at a concentration of 300 mM, all captured antibody was eluted at pH 6.5, meaning a reduction in acidity by a factor of 10 (Fig. 3B). At pH 7, a fraction of less than half of the antibody sample remained on the column. Despite the fact that all bound antibody could be eluted at pH 6.5, the eluate was rather dilute, whereas a considerably smaller difference could be observed between pH 6 and 5.5. For large scale production, elution at pH 6.5 could be a suitable alternative for IgG1 mAbs that are unstable at lower pH, but to obtain a concentrated product comparable to acidic elution on commercial Protein A matrices, sodium chloride should be used at pH 6 or 5.5. Elution at pH 6 still represents more than a 3-fold increase in pH compared to elution at pH 5.5, and thus notably milder conditions for the antibodies.

To investigate what concentration of sodium chloride that is required for efficient elution of trastuzumab at pH 5.5, lower concentrations were tested. Complete elution can be achieved by merely 50 mM sodium chloride (Fig. 3C), confirmed by a control acid elution afterwards showing no residual antibody on the column (data not shown). This is a lower concentration than needed when using EDTA as eluent [17]. At a physiological concentration of 150 mM sodium chloride, a sharp elution peak can be obtained, very similar to that of 300 mM sodium chloride. Taken together, sodium chloride can replace EDTA for equal elution of IgG1 antibodies from Z_{Ca} TetraCys, at similar concentrations observed for EDTA, with milder elution pH and resulting in a very concentrated eluate. A previously published study involving loop engineering aiming to destabilize the Z domain also enabled a milder elution pH, but only by an increase to pH 4.5, and rendered a less concentrated eluate compared to Z_{Ca} TetraCys [13]. The lowest sodium chloride concentration that is proven successful for elution from Z_{Ca} TetraCys is 30 times lower than the reported concentration used for elution from the thermo-responsive Protein A-based ligand Byzen Pro, which has been shown to elute antibodies using 1.5 M sodium chloride

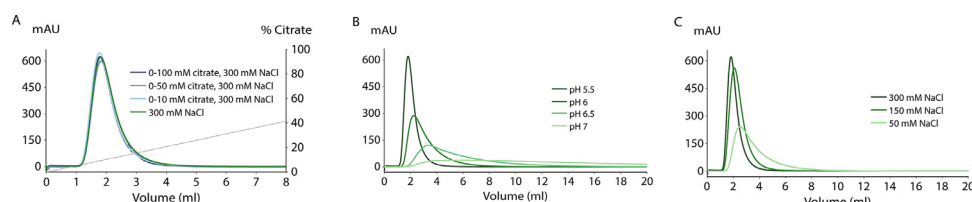


Fig. 3. Optimization of sodium chloride-containing buffers for elution of trastuzumab from Z_{Ca} TetraCys. (A) Overlay of gradient elution chromatograms with decreasing final citrate concentrations and a constant sodium chloride concentration throughout the gradients. Trastuzumab injected onto Z_{Ca} TetraCys was eluted with 300 mM sodium chloride and gradients (gray) of up to 10 mM (pale blue), 50 mM (light blue), and 100 mM (dark blue) citrate or no citrate at all (green), at pH 5.5. Elution with only sodium chloride was as efficient as the combination of sodium chloride and citrate for elution of the bound antibody. (B) Overlay of isocratic elution chromatograms at 300 mM sodium chloride and different pH (5.5 in dark green, 6 in mid green, 6.5 in light green and 7 in pale green). Elution at pH 6.5 was sufficient to elute all captured trastuzumab, but lower pH is preferable to obtain a concentrated eluate. (C) Overlay of isocratic chromatograms demonstrating the elution profiles at decreasing sodium chloride concentrations of 300 mM (dark green), 150 mM (mid green) and 50 mM (pale green), at pH 5.5. A concentration of 50 mM sodium chloride was sufficient for complete trastuzumab elution.

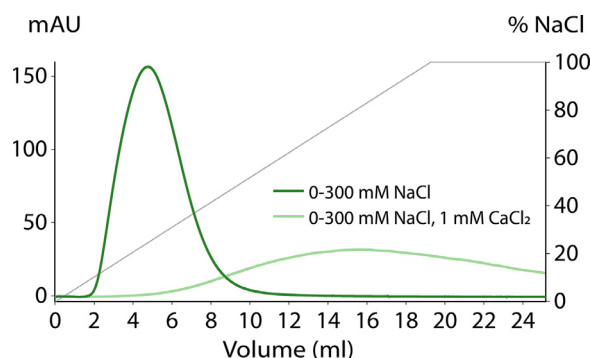


Fig. 4. Calcium-dependent elution with sodium chloride. Trastuzumab was eluted from Z_{Ca}TetraCys with a gradient (gray) of 0–300 mM sodium chloride at pH 5.5, with (pale green) or without (mid green) 1 mM CaCl₂. The large difference in elution profile in the presence and absence of calcium demonstrates the calcium-dependency of the sodium chloride-mediated elution.

[15]. The use of such a high salt concentration can have a large effect on the following steps of the mAb downstream process, such as ion-exchange chromatography, which could require dilution of the eluate to a very large product volume.

Since sodium chloride-mediated elution can be performed at higher pH than EDTA-mediated elution, the mechanism of elution might differ and is likely multiplex. To investigate if this mechanism is calcium-dependent, a sodium chloride gradient was applied for elution of trastuzumab from Z_{Ca}TetraCys, both in the presence and absence of calcium in the elution buffer. A substantial difference between the two elution chromatograms could be observed (Fig. 4). The addition of calcium at a low concentration resulted in a very delayed, minimal elution peak with a large amount of antibody retained on the column. This confirms that sodium chloride not only interferes with the antibody binding site but also, and most importantly, with the calcium binding of the Z_{Ca}TetraCys loop. The calcium-binding loop of Z_{Ca}TetraCys releases calcium at an increased ionic strength and a slightly acidic pH, and the release of calcium subsequently leads to the dissociation of the bound antibody (mid green), according to the same principle as for the chelators. This was shown to be largely counteracted by the constant supply of calcium throughout the elution (pale green). Consequently, sodium chloride contributes to the elution of antibodies from Z_{Ca}TetraCys in a calcium-dependent manner.

However, as mentioned, there is an additional factor promoting the antibody release: the pH. As previously described, Z_{Ca} is most likely affected by an altered calcium affinity at pH below neutral conditions, leading to a more rapid and complete antibody dissociation [16]. Of the residues coordinating calcium in the Z_{Ca} loop (Asp, Asn, Asp, Glu and Tyr), aspartate and glutamate have side chain pK_a values close to pH 5.5, depending on the surrounding amino acids in the three-dimensional structure. This means that lowering of the pH will cause a shift in the amino acid charges, increasing the protonation of these to different degrees given the adjacent amino acids. The increased positive charge of the loop might facilitate the release of the positively charged calcium ion from the loop. This reduced affinity for calcium at slightly acidic pH could thus increase the chances to release the calcium ion from the loop. Yet, when eluting trastuzumab with sodium chloride at pH 7, most antibody was evidently released, further validating a sodium chloride-mediated calcium-dependent antibody release rather than pH-dependent. The affinity for calcium appears to be decreased by the increased ionic strength of the elution buffer; the large excess of negatively charged chloride ions in relation to calcium bound by the loop attracts the calcium ions and may draw them away from the loop. It is possible that

other negatively charged ions would give similar results, but considering the low concentrations of sodium chloride required for this cost-efficient elution and the fact that it is also used in the wash buffer, including other salts would likely not provide a significant advantage in this process.

3.3. Different subclasses of IgG require different pH for sodium chloride-mediated elution

Besides the IgG1 subclass, which most antibody therapeutics are based on and which trastuzumab belongs to, Protein A also binds to IgG2 and IgG4. Z_{Ca}TetraCys has previously been shown to display varying elution characteristics for these different subclasses [17]. To elucidate the differences with sodium chloride as eluent, the optimal elution conditions for mild elution of each antibody subclass were evaluated. Both sodium chloride concentration and pH of the elution buffer were taken into consideration, and IgG4 could be eluted into a concentrated product at neutral pH with only 150 mM sodium chloride (Fig. 5A). For IgG2, an increase in sodium chloride concentration to 300 mM was needed to achieve an equal elution at neutral pH. Since all captured monoclonal IgG1 could be eluted with sodium chloride at pH 6, and a concentration of 150 mM resulted in a similar elution peak to 300 mM at pH 5.5, 150 mM sodium chloride was evaluated at pH 6. At these conditions, IgG1 eluted into an equally concentrated eluate as the other two subclasses. In fact, the ratio between the peak widths for the two concentrations at pH 6 corresponded to the same ratio at pH 5.5 (Table S2).

To further understand the interaction between Z_{Ca}TetraCys and each IgG subclass, the affinities in the presence of calcium were determined by SPR analysis. With the monoclonal antibodies immobilized on separate surfaces, the Z_{Ca}TetraCys ligand was injected as the analyte, resulting in dissimilar sensorgrams as well as affinities for the different subclasses (Fig. 5B). The dissociation equilibrium constant (K_D) of Z_{Ca}TetraCys to IgG1 was estimated to 90 nM. The affinity for the tetrameric ligand is consequently about four times higher than previously determined for the monomeric Z_{Ca} [16], which could probably be attributed to avidity effects. It is also distinctly higher than the affinities of Z_{Ca}TetraCys to IgG2 and IgG4 with 690 nM and 780 nM, respectively. This is reflected in the higher binding signal and slower dissociation for IgG1. Conversely, IgG4, with the lowest affinity, displayed a slightly lower signal and faster dissociation compared to IgG2, confirming the smaller difference in affinity between these two subclasses. This difference in binding of Z_{Ca}TetraCys to IgG1, IgG2, and IgG4 could explain the need for harsher elution conditions at pH 6 for efficient dissociation of IgG1 from Z_{Ca}TetraCys. Additionally, IgG2 required a slightly higher sodium chloride concentration than IgG4 for elution at pH 7, likely caused by the somewhat higher affinity in comparison to IgG4. The two latter subclasses have both been reported to be more susceptible to aggregation [21–23], and are therefore especially benefitted by the neutral elution conditions. Also, more and more therapeutic antibodies are being based on IgG2 and IgG4, now constituting around half of the antibodies under regulatory review [1].

3.4. Sodium chloride-mediated elution yields high purity and antibody recovery

To obtain a pure protein with minimal loss, selectivity and recovery are important factors of an affinity chromatography resin. Therefore, the subclasses with the highest and lowest affinity for Z_{Ca}TetraCys, IgG1 and IgG4 respectively, were further examined in a true purification setup. Capture of IgG1 and IgG4 from Chinese hamster ovary (CHO) cell supernatants was followed by mild elution of IgG1 at pH 5.5 and IgG4 at pH 7. During sample appli-

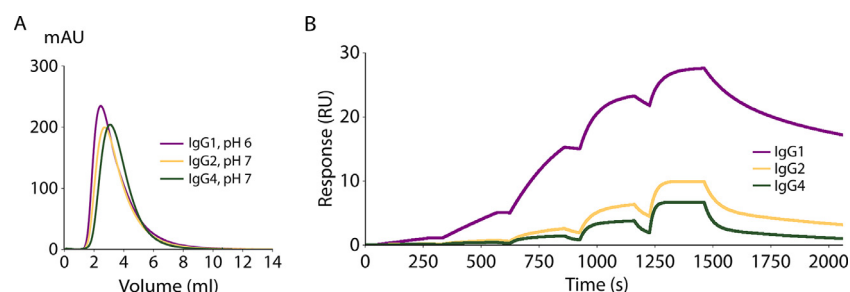


Fig. 5. Analysis of the binding characteristics of $Z_{\alpha}TetraCys$ to different subclasses of IgG. (A) Chromatograms of mild elution of monoclonal IgG1 (purple), IgG2 (yellow), and IgG4 (green) from $Z_{\alpha}TetraCys$ with 150 mM NaCl, pH 6 (IgG1), 300 mM NaCl, pH 7 (IgG2), and 150 mM NaCl, pH 7 (IgG4). (B) SPR sensorgrams displaying the binding of $Z_{\alpha}TetraCys$ to IgG1 (purple), IgG2 (yellow) and IgG4 (green) using single-cycle kinetics. IgG1 exhibited the highest binding signal and the slowest dissociation, while IgG4 displayed the fastest dissociation and lowest signal. This is consistent with the conditions required for elution of each subclass from $Z_{\alpha}TetraCys$.

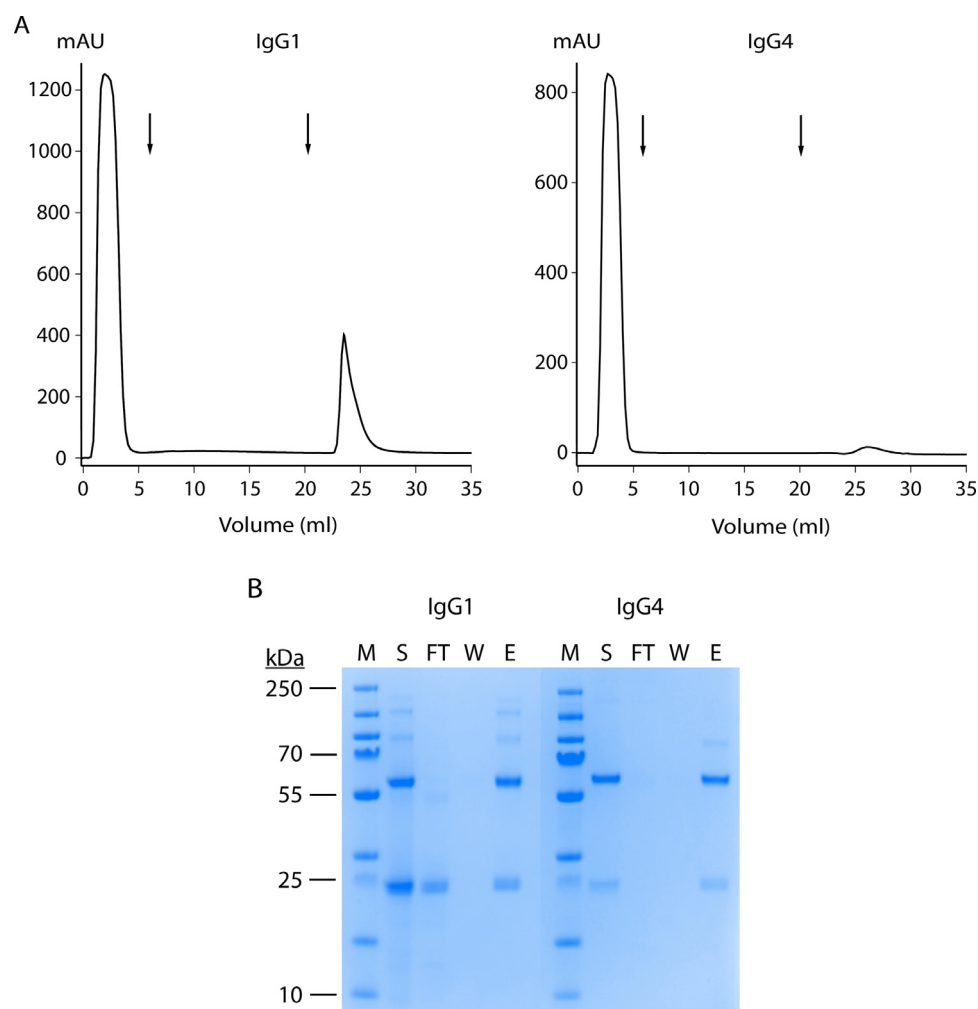


Fig. 6. Purification of CHO cell culture supernatant containing recombinantly produced IgG1 (trastuzumab) at around 0.8 g/L, or spiked with IgG4 to approximately 0.1 g/L, on $Z_{\alpha}TetraCys$. (A) Chromatograms displaying the purification procedures, including sodium chloride-mediated elution of the captured IgG1 and IgG4 at pH 5.5 and pH 7, respectively. Arrows indicate the transition from sample application to wash and from wash to elution, from left to right. (B) SDS-PAGE displaying equal fractions of the supernatant (S), the flow through (FT) during sample application, the wash (W), and the eluate (E) from the purifications of IgG1 and IgG4. A molecular weight marker (M) is included with bands at 10, 15, 25, 35, 55, 70, 100, 130 and 250 kDa. No antibody was detected in the flow through or wash, and the same amount of antibody that was present in the supernatant was found in the eluate.

cation, high UV signals were measured, followed by signals adjacent to baseline during washing, and concentrated elution peaks (Fig. 6A). Sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) analysis shows impurities present in the supernatant as well as the flow through, most notably for IgG1, corresponding to the large peak during sample application (Fig. 6B).

It also confirms that the antibody was not released during sample application nor during washing with sodium chloride with calcium present. For IgG1, the band present in the flow through of the same size as the antibody light chain is most likely overexpressed light chain during production, that lacks the heavy counterpart. This band is also more intense than the heavy chain in the

supernatant, and is considered an impurity that was removed in virtue of Z_{Ca} TetraCys' specific binding to the antibody heavy chain. Pure antibody was detected in the eluate, in equal amounts to the supernatant for both IgG1 and IgG4, demonstrating high antibody recovery of the purifications. Accordingly, Z_{Ca} TetraCys enables the selective capture of full-length antibodies as well as the possibility to perform washing with sodium chloride, despite the sodium chloride-dependent elution properties. However, in order to use sodium chloride in the wash buffer, where it is typically included for efficient removal of impurities caught on the Protein A column prior to elution of the antibodies, calcium needs to be added. That way, the antibody was not released during washing with a Tris buffer containing calcium and 150 mM sodium chloride at pH 7.5, when captured from CHO cell supernatant. The calcium-dependent properties of Z_{Ca} TetraCys can thus be utilized to prevent sodium chloride-mediated premature elution during washing, as opposed to for other sodium chloride-sensitive resins, such as Byzen Pro [15].

To further investigate the recovery that can be obtained when purifying the subclass with the highest and lowest affinity for Z_{Ca} TetraCys, IgG1 and IgG4, pure antibody was captured and eluted in the same way as for the CHO supernatant purifications. High antibody recovery of 99% for IgG1 (rituximab) and 88% for IgG4 was achieved. IgG2 can be expected to be recovered at rates in between these if considering the difference in affinities. The remaining antibody could be detected by absorbance measurements in the flow through (4%) and wash (8%) for IgG4, whereas no antibody was released during washing for IgG1, but a negligible amount (<1%) during sample application. All antibody was thus recovered during the purification, before regeneration of the column. SDS-PAGE analysis of each step of the purification procedure visualizes the small amount of antibody in the flow through and wash for IgG4, and displays bands of very similar intensities for the pure samples loaded onto the column and the eluates (Fig. S2). The IgG4 antibody detected in the flow through and wash indicates that the weaker affinity was sufficient to retain almost all antibody on Z_{Ca} TetraCys, but not everything. This assumption is supported by the difference in conductivity between the wash buffer and the neutral pH elution buffer used for IgG4, which is smaller than for IgG1 with its lower pH elution buffer, implicating more similar conditions in the IgG4 purification procedure. To prevent the release of antibody in the flow through and wash and thereby improve the recovery, the conductivity could be decreased by lowering the sodium chloride concentration of the wash buffer. Moreover, the conductivity of the CHO cell supernatant is considerably lower than the wash buffer, in which the pure antibody was loaded, meaning a higher recovery can likely be obtained when the antibody is captured in a true purification setup. All things considered, great selectivity and recovery can be obtained with sodium-chloride mediated elution from Z_{Ca} TetraCys. Additional key aspects of a Protein A resin include the stability of the ligand itself and the degree of ligand leaching, which could be further looked into. A promising first indication is the consistent performance of Z_{Ca} TetraCys throughout numerous purifications on the same column. However, Z_{Ca} TetraCys has not been optimized to tolerate alkaline treatment, which is widely used for cleaning of chromatography resins [11], in contrast to resins like MabSelect SuRe [24] and the new MabSelect Prisma. Evaluation and possible improvement of the ligand stability under alkaline conditions will be addressed in future studies. When applying Z_{Ca} TetraCys in a biotherapeutic production process, the full advantage of the resin can only be utilized when incorporating a mild virus inactivation step avoiding low pH. There are already up-and-coming alternatives to the acidic virus inactivation downstream in the purification train, for example solvent/detergent treatment [25–28].

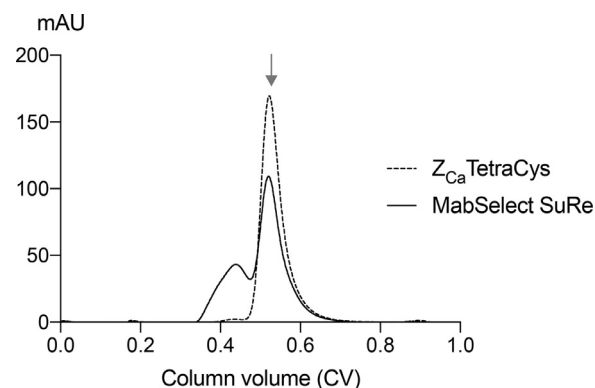


Fig. 7. SEC chromatograms of an IgG4 antibody post purification on Z_{Ca} TetraCys (dashed) and MabSelect SuRe (solid). The antibody exposed to mild elution conditions on Z_{Ca} TetraCys demonstrates <1% high-molecular weight species while this constituted 34% of all antibody eluted by low pH from MabSelect SuRe. A reference protein of 158 kDa was included which is indicated by the arrow.

3.5. Mild elution prevents the formation of high-molecular weight species

The purpose of Z_{Ca} TetraCys is to allow for the purification of acid-sensitive antibodies and Fc-fusion proteins that cannot withstand the current acidic conditions used for elution from Protein A resins. As described, a consequence of the acidic exposure is antibody aggregation [8–10], which can lead to the loss of large amounts of antibody product, while mild elution close to neutral pH should avert this problem. To investigate this, SEC analysis was conducted on the eluates obtained from capture and elution of an IgG4 antibody on Z_{Ca} TetraCys and MabSelect SuRe. For MabSelect SuRe, where pH 3 was used to elute the antibody, two large peaks were detected (Fig. 7). After elution from Z_{Ca} TetraCys at pH 7, only one peak appeared, at the same retention time as the 158 kDa reference protein. The first peak eluted from the SEC column when analyzing the MabSelect SuRe eluate, representing high-molecular weight species, constituted 34% of all eluted antibody, whereas this fraction was less than 1% for Z_{Ca} TetraCys. By eluting an IgG4 antibody at neutral pH from Z_{Ca} TetraCys, the proportion of high-molecular weight species in the eluate could thus be reduced from more than one third of all eluted antibody for MabSelect SuRe down to a minor fraction. The difference in elution pH used to obtain elution of all captured antibody from both matrices consequently had a major impact on aggregate formation, which was close to fully avoided at neutral pH with Z_{Ca} TetraCys. Besides this specific antibody, there are other antibodies that are more tolerant or less tolerant of low pH, which means that the effect of mild purification using Z_{Ca} TetraCys will be dependent on the purified antibody. In addition to analyzing the impact on aggregation, the elution pH has also been shown to affect the function of purified antibodies by purification of polyclonal antibodies from human serum, which exhibited a higher affinity for an antigen in SPR when purified on Z_{Ca} TetraCys compared to MabSelect SuRe (data not shown). Regarding antibody functionality, we foresee that Z_{Ca} TetraCys will have a similar impact as traditional Protein A-based matrices. That is based on the fact that Z_{Ca} TetraCys is very similar to the parental Z domain of Protein A, which the MabSelect SuRe and Byzen Pro resins are also based on [11,24], and interacts with Fc by helix one and two (Fig. 1) in a similar way as the Z domain [16]. However, this should be investigated further in upcoming studies. As demonstrated, this novel ligand is of great significance for the purification of less stable antibodies without compromising the quality and yield. The large difference in yield of functional, non-aggregated IgG4 antibody between Z_{Ca} TetraCys and

MabSelect SuRe means that Z_{Ca}TetraCys can compete with commercial matrices optimized for high binding capacities and recovery rates.

4. Conclusions

In this study we concluded that EDTA is favorably replaceable for mild antibody elution from a Z_{Ca}TetraCys matrix by both citrate- and sodium chloride-mediated elution, for a more environmentally friendly process. Sodium chloride should be used at a low concentration of 150 mM for efficient elution of IgG1 at pH 6, IgG4 at pH 7 and at 300 mM for elution of IgG2 at pH 7 into a highly concentrated eluate. Thus, we present a chromatography ligand with revolutionary features by combining the selectivity of Protein A with antibody binding properties that can be modulated by sodium chloride. Yet, the calcium-dependent properties of Z_{Ca}TetraCys allow for efficient conventional washing using sodium chloride in the presence of calcium. Antibody at exceptional purity and with high recovery can be obtained from the high-capacity Z_{Ca}TetraCys resin by mild, cheap and eco-friendly elution in a similar manner to ion-exchange chromatography. The mild conditions were proven to considerably influence the formation of antibody aggregates, which were negligible in comparison to the large proportion of high-molecular weight species obtained from MabSelect SuRe. This novel purification strategy opens up the possibility for the development of a wider range of therapeutic antibodies and Fc-fused proteins, regardless of poor stability at low pH.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

SH has filed a patent application regarding the novel Z_{Ca} domain.

CRediT authorship contribution statement

Julia Scheffel: Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Sophia Hober:** Conceptualization, Supervision, Writing - review & editing, Project administration, Funding acquisition.

Acknowledgements

This work is financially supported by Vinnova (the AdBIOPRO excellence center) under Grant 2016-05181 and the Swedish research council under Grant 2016-04717. The authors would like to thank Björn Norén at Cytiva for the directed coupling of the purification ligands on a matrix. The CHO cell supernatant containing trastuzumab was also provided by Cytiva. The authors want to extend their gratitude to Alois Jungbauer's lab at the Department of Biotechnology, University of Natural Resources and Life Sciences, Vienna and BioInvent for kindly sharing monoclonal IgG2 and IgG4, respectively. Sara Kanje is acknowledged for the development of the Z_{Ca} domain and for fruitful discussions regarding this publication, and Jesper Borin for assisting with purifications.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.chroma.2020.461843](https://doi.org/10.1016/j.chroma.2020.461843).

References

- [1] H Kaplon, JM. Reichert, Antibodies to watch in 2019, *MAbs* 11 (2) (2019) 219–238, doi:[10.1080/19420862.2018.1556465](https://doi.org/10.1080/19420862.2018.1556465).

- [2] AC Roque, CS Silva, MA. Taipa, Affinity-based methodologies and ligands for antibody purification: advances and perspectives, *J Chromatogr A* 1160 (1–2) (2007) 44–55, doi:[10.1016/j.chroma.2007.05.109](https://doi.org/10.1016/j.chroma.2007.05.109).
- [3] A Tscheliessnig, P Satzer, N Hammerschmidt, H Schulz, B Helk, A. Jungbauer, Ethanol precipitation for purification of recombinant antibodies, *J Biotechnol* 188 (2014) 17–28, doi:[10.1016/j.jbiotec.2014.07.436](https://doi.org/10.1016/j.jbiotec.2014.07.436).
- [4] LN Mao, JK Rogers, M Westoby, L Conley, J. Pieracci, Downstream antibody purification using aqueous two-phase extraction, *Biotechnol Prog* 26 (6) (2010) 1662–1670, doi:[10.1002/btpr.477](https://doi.org/10.1002/btpr.477).
- [5] P. Gagnon, Technology trends in antibody purification, *J Chromatogr A* 1221 (2012) 57–70, doi:[10.1016/j.chroma.2011.10.034](https://doi.org/10.1016/j.chroma.2011.10.034).
- [6] AJ Paul, K Schwab, F. Hesse, Direct analysis of mAb aggregates in mammalian cell culture supernatant, *BMC Biotechnol* 14 (2014) 99, doi:[10.1186/s12896-014-0099-3](https://doi.org/10.1186/s12896-014-0099-3).
- [7] M Vázquez-Rey, Lang DA, Aggregates in monoclonal antibody manufacturing processes, *Biotechnol Bioeng* 108 (2011) 1494–1508, doi:[10.1002/bit.23155](https://doi.org/10.1002/bit.23155).
- [8] AA Shukla, P Gupta, X. Han, Protein aggregation kinetics during Protein A chromatography. Case study for an Fc fusion protein, *J Chromatogr A* 1171 (2007) 22–28, doi:[10.1016/j.chroma.2007.09.040](https://doi.org/10.1016/j.chroma.2007.09.040).
- [9] B Liu, H Guo, J Xu, T Qin, L Xu, J Zhang, Q Guo, D Zhang, W Qian, B Li, et al., Acid-induced aggregation propensity of nivolumab is dependent on the Fc, *MAbs* 8 (2016) 1107–1117, doi:[10.1080/19420862.2016.1196521](https://doi.org/10.1080/19420862.2016.1196521).
- [10] AR Mazzer, X Perraud, J O'Hara, DG Bracewell, Protein A chromatography increases monoclonal antibody aggregation rate during subsequent low pH virus inactivation hold, *J Chromatogr A* 1415 (2015) 83–90, doi:[10.1016/j.chroma.2015.08.068](https://doi.org/10.1016/j.chroma.2015.08.068).
- [11] S Kanje, J Scheffel, J Nilvebrant, S. Hober, Chapter 2 - Engineering of Protein A for improved purification of antibodies and Fc-fused proteins, in: A Matte (Ed.), *Approaches to the Purification, Analysis and Characterization of Antibody-Based Therapeutics*, Elsevier, 2020, pp. 35–54, doi:[10.1016/B978-0-08-103019-6.00002-3](https://doi.org/10.1016/B978-0-08-103019-6.00002-3).
- [12] TM Pabst, R Palmgren, A Forss, J Vasic, M Fonseca, C Thompson, WK Wang, X Wang, AK. Hunter, Engineering of novel Staphylococcal Protein A ligands to enable milder elution pH and high dynamic binding capacity, *J Chromatogr A* 1362 (2014) 180–185, doi:[10.1016/j.chroma.2014.08.046](https://doi.org/10.1016/j.chroma.2014.08.046).
- [13] S Gulich, M Uhlen, S. Hober, Protein engineering of an IgG-binding domain allows milder elution conditions during affinity chromatography, *J Biotechnol* 76 (2–3) (2000) 233–244, doi:[10.1016/S0168-1656\(99\)00197-2](https://doi.org/10.1016/S0168-1656(99)00197-2).
- [14] I Koguma, S Yamashita, S Sato, K Okuyama, Y. Katakura, Novel purification method of human immunoglobulin by using a thermo-responsive protein A, *J Chromatogr A* 1305 (2013) 149–153, doi:[10.1016/j.chroma.2013.07.015](https://doi.org/10.1016/j.chroma.2013.07.015).
- [15] W Krepper, P Satzer, BM Beyer, A. Jungbauer, Temperature dependence of antibody adsorption in protein A affinity chromatography, *J Chromatogr A* 1551 (2018) 59–68, doi:[10.1016/j.chroma.2018.03.059](https://doi.org/10.1016/j.chroma.2018.03.059).
- [16] S Kanje, R Venskutonytė, J Scheffel, J Nilvebrant, K Lindkvist-Petersson, S. Hober, Protein engineering allows for mild affinity-based elution of therapeutic antibodies, *J Mol Biol* 430 (2018) 3427–3438, doi:[10.1016/j.jmb.2018.06.004](https://doi.org/10.1016/j.jmb.2018.06.004).
- [17] J Scheffel, S Kanje, J Borin, S. Hober, Optimization of a calcium-dependent Protein A-derived domain for mild antibody purification, *MAbs* 11 (8) (2019) 1492–1501, doi:[10.1080/19420862.2019.1662690](https://doi.org/10.1080/19420862.2019.1662690).
- [18] P Bajpai, 7 - Chemicals Used in Deinking and Their Function, in: P Bajpai (Ed.), *Recycling and Deinking of Recovered Paper*, Elsevier, Oxford, 2014, pp. 121–137, doi:[10.1016/B978-0-12-416998-2.00007-6](https://doi.org/10.1016/B978-0-12-416998-2.00007-6) page.
- [19] B. Nörtemann, Biodegradation of EDTA, *Appl Microbiol Biotechnol* 51 (1999) 751–759, doi:[10.1007/s002530051458](https://doi.org/10.1007/s002530051458).
- [20] E Keowmaneechai, DJ. McClements, Influence of EDTA and citrate on physicochemical properties of whey protein-stabilized oil-in-water emulsions containing CaCl₂, *J Agric Food Chem* 50 (24) (2002) 7145–7153, doi:[10.1021/jf020489a](https://doi.org/10.1021/jf020489a).
- [21] B Liu, H Guo, J Xu, et al., Acid-induced aggregation propensity of nivolumab is dependent on the Fc, *MAbs* 8 (6) (2016) 1107–1117, doi:[10.1080/19420862.2016.1197443](https://doi.org/10.1080/19420862.2016.1197443).
- [22] SB Hari, H Lau, VI Razinkov, S Chen, RF. Latypov, Acid-induced aggregation of human monoclonal IgG1 and IgG2: molecular mechanism and the effect of solution composition, *Biochemistry* 49 (43) (2010) 9328–9338, doi:[10.1021/bi100841u](https://doi.org/10.1021/bi100841u).
- [23] T Ito, K. Tsumoto, Effects of subclass change on the structural stability of chimeric, humanized, and human antibodies under thermal stress, *Protein Sci* 22 (11) (2013) 1542–1551, doi:[10.1002/pro.2340](https://doi.org/10.1002/pro.2340).
- [24] M Linholt, S Gulich, T Gräslund, A Simon, M Karlsson, A Sjöberg, K Nord, S. Hober, Improving the tolerance of a protein A analogue to repeated alkaline exposures using a bypass mutagenesis approach, *Proteins* 55 (2) (2004) 407–416, doi:[10.1002/prot.10616](https://doi.org/10.1002/prot.10616).
- [25] Nur I, Bar L Method of the inactivation of viruses by a solvent-detergent combination and nanofiltration, (US6468733B2). 2002.
- [26] PL Roberts, Virus inactivation by solvent/detergent treatment using Triton X-100 in a high purity factor VIII, *Biologicals* 36 (2008) 330–335, doi:[10.1016/j.biologics.2008.06.003](https://doi.org/10.1016/j.biologics.2008.06.003).
- [27] PL Roberts, D Lloyd, PJ. Marshall, Virus inactivation in a factor VIII/VWF concentrate treated using a solvent/detergent procedure based on polysorbate 20, *Biologicals* 37 (2009) 26–31, doi:[10.1016/j.biologics.2009.05.003](https://doi.org/10.1016/j.biologics.2009.05.003).
- [28] DL Martins, J Sencar, N Hammerschmidt, B Tille, J Kinderman, TR Kreil, A. Jungbauer, Continuous Solvent/Detergent Virus Inactivation Using a Packed-Bed Reactor, *Biotechnology Journal* 14 (8) (2019) 1800646, doi:[10.1002/biot.201800646](https://doi.org/10.1002/biot.201800646).